Note 4:- If the given year is like L.Y+3 (or) L.Y-1 then same calendar year after 11 years.

* Note 5:- If the given year is like [Non-leap century year - 28] years ~~then same~~ which is leap year then same calendar year is not after 28 years. but after 40 years same calendar year will come.

Ex:9:- What will be the same calendar year of 2021?

M-1:-

1	1	1	2	1	1	1	1	1.
2021	22	23	24	25	26	27	28	29 . . .

always after 7 multiple number of days same day is repeating.

∴ from 2021 → 1 odd day to 2026 → 1 odd day the odd days count is '7' then next year 2027 starts with same day. It ~~year~~ is same calendar year. if given year is N.L.Y/L.Y then it's also N.L.Y/L.Y. if not go through the same counting of odd days till to get same calendar year. So 2027 is same year.

M-2:- 01-01-2021 → 5 odd days ⇒ Friday.

01-01-2022 → Saturday (+1) → odd day in 2022

01-01-2023 → Sunday (+1)

01-01-2024 → Monday (+1)

01-01-2025 → +2 → ~~Wednesday~~ Tuesday

01-01-2026 → ~~Thursday~~ Thursday (+1)

01-01-2027 → Friday So, 2027

Mensuration

[Vector India Pvt. Ltd]

By Sunil Guptha Yarramalla.

↳ Mensuration is a part of geometry which deals with the closed figures.

↳ Mensuration / closed figures are classified into 2 types based on dimension.

(i) 2D figures (ii) 3D figures.

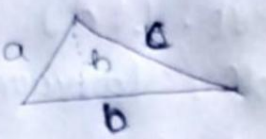
(i) 2D figures:- Again importantly 2D figures are classified as: 1) Δ'les
2) Quadrilaterals
3) Circle & semi-Circle
4) Polygons. [Not required here]

1) Δ'les:- Based on all properties of Δ'les like based on angle & lengths of sides Δ'les categorised into to 5 types.

(i) Scalene Δ'le
(ii) Equilateral Δ'le
(iii) Isosceles Δ'le.
(iv) Right angle Δ'le
(v) Right angled Isosceles Δ'le

→ Basic properties of any Δ'le is 1) Δ'le is a 3 sided closed figure.
② Sum of angles is 180° ③ Sum of 2 angles $\geq$ 3rd angle.
④ Height (or) altitude nothing but Δ'les Bisector drawn from a vertex to its opposite side.
⑤ Sum of two sides $\geq$ 3rd side [In terms of lengths]

1) Scalene Δle:- All sides are different & all angles are different.



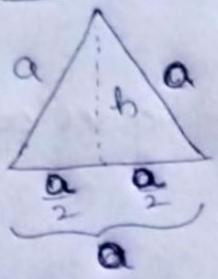
Area = $\frac{1}{2}$ × base × height

$= \frac{1}{2}bh$

Semiperimeter = $\frac{a+b+c}{2} = s$ $\quad P = a+b+c$

Area on semi-perimeter = $\sqrt{s(s-a)(s-b)(s-c)}$

2) Equilateral Δle:- All angles are equal & all sides equal



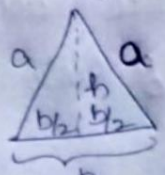
height = $h = \sqrt{a^2 - (\frac{a}{2})^2}$ [by pythagoreous theorem]

$= \sqrt{\frac{4a^2 - a^2}{4}} = \frac{\sqrt{3}a}{2}$

$A = \frac{1}{2} \times a \times \frac{\sqrt{3}a}{2} = \frac{\sqrt{3}a^2}{4}$, $P = 3a$

$A_S = \sqrt{s(s-a)(s-a)(s-a)} = (s-a)\sqrt{s(s-a)}$

3) Isosceles Δle:- 2 sides are equal 3rd side is different (or) 2 angles are equal & 3rd angle is different

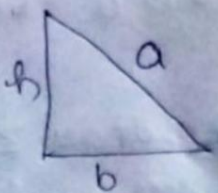


height = $h = \sqrt{a^2 - (\frac{b}{2})^2} = \sqrt{\frac{4a^2 - b^2}{4}} = \frac{\sqrt{4a^2 - b^2}}{2}$

$A = \frac{1}{2} \times b \times \frac{\sqrt{4a^2 - b^2}}{2} = \frac{b\sqrt{4a^2 - b^2}}{4}$, $P = 2a + b$

$A_S = \sqrt{s(s-a)(s-a)(s-b)} = (s-a)\sqrt{s(s-b)}$

4) Right angle Δle:- All sides are different and one angle must be 90° (or) one angle is right angles to each other with the other 2 sides/angles.

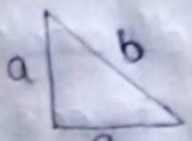


$A = \frac{1}{2} \times b \times h$, $P = a+b+h$

$A_S = \sqrt{s(s-a)(s-b)(s-h)}$.

5) Right angled Isosceles Δle:- two sides are equal & 3rd side is different, one angle is 90 & the other two angles sum is 90 & equal = 45°.

(139)



$A = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times a \times a$

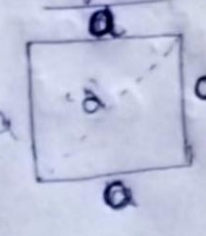
$= \frac{1}{2}a^2$, $P = a+a+b = 2a+b$

$A_S = \sqrt{s(s-a)(s-a)(s-b)} = (s-a)\sqrt{s(s-b)}$

(2) Quadrilaterals :- General properties of Quadrilaterals are (i) 4 sided closed figure. (ii) Sum of angles 360° (iii) forms 2 diagonals. (iv) Area can be formed by making pt onto 2 Δles.

Quadrilaterals are mainly 5 types:-

(i) Square :- All sides are equal & all angles are equal = 90°. diagonals are also equal.



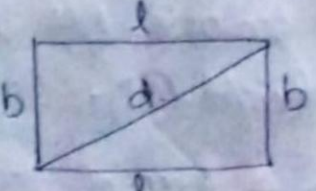
(ii) By deviding pt with diagonal pt is devided onto 2 equal right angles isosceles Δles

Area. = 2 [Area of Right angles Isosceles Δle]

$= 2\,\frac{1}{2}a^2 = \boxed{a^2}$

$P = 4a$, $d = \sqrt{a^2+a^2} = \sqrt{2}\,a$

(ii) Rectangle :- Opposite sides are equal & ||el & all angles are equal = 90°. with diagonal pt is devided onto 2 equal right angle Δles.



Area = 2 [Area of right angle Δle]

$= 2\left[\frac{1}{2}lb\right] = lb$

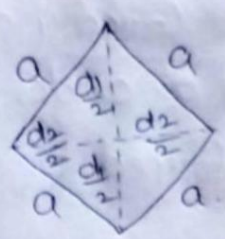
$P = 2(l+b)$

$d = \sqrt{l^2+b^2}$ [By Pythogerous theorem]

$d^2 = l^2+b^2 \Rightarrow l^2 = d^2-b^2 \Rightarrow l = \sqrt{d^2-b^2}$

$\Rightarrow b = \sqrt{d^2-l^2}$

3) Rhombus:- All sides are equal but not all angles. Opposite angles are equal. two diagonals bisects it into 4 equal right angle Δles.



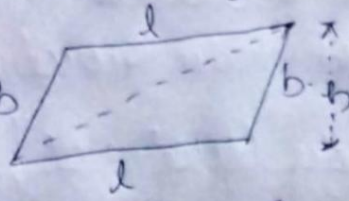
Area = 4 [Area of each right angle Δle]

$= 4\left[\frac{1}{2}\frac{d_1}{2}\cdot\frac{d_2}{2}\right] = 4\left[\frac{d_1 d_2}{8_2}\right]$

$\Rightarrow$ Area $= \frac{1}{2} d_1 d_2$

$$\text{Side} = a = \sqrt{\left(\frac{d_1}{2}\right)^2 + \left(\frac{d_2}{2}\right)^2}$$

4) Parallellogram:- Opposite sides are equal & ||el. Opposite angles are equal. but not 90°. diagonal bisects it into 2 equal scalare Δles.

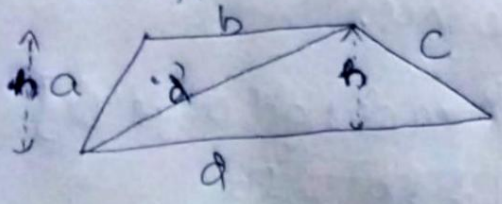


A = 2 [Area of scalare Δle] = 2 [$\frac{1}{2}$ base × height]

= base × height.

$P = 2(l+b)$.

5) Trapeziums:- All sides are different and one pair of opposite sides are ||el.



Diagonal bisects it into 2 diff-erent scalare Δles with same height (h).

Area = Area of 1st Δle with base 'b' + Area of 2nd Δle with base 'd'.

$= \frac{1}{2}bh + \frac{1}{2}dh = \frac{1}{2}h.[b+d]$